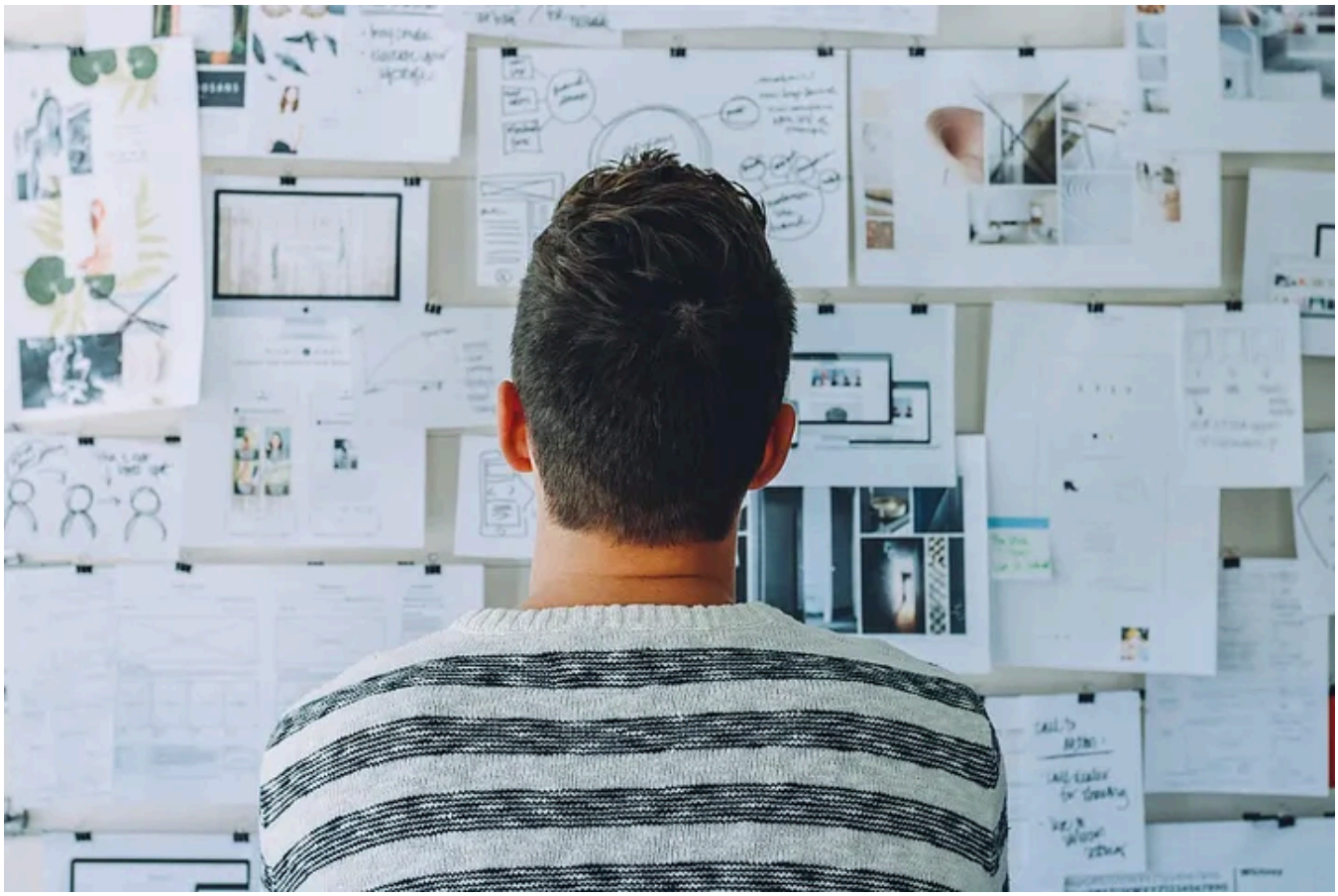
AI Writes Code. You Write the Specs.

Specifications are all you need

3 min read · Just now

Yogesh Haribhau Kulkarni (PhD)

Listen Share ••• More



(Source: Pixabay)

Though there is huge hype around AI writing code on its own, the power of vibe coding, and autonomous agents running for weeks to prepare full systems, the reality is far from that. The breakthrough is not autonomous coding. The breakthrough is in specification.

AI is now a super coder. It can generate thousands of lines of working code in minutes. It can scaffold applications, integrate APIs, write unit tests, and refactor legacy systems. In some environments, AI already writes the majority of new code.

But here is the catch. AI does exactly what you ask. If your instructions are vague, incomplete, or misaligned with business needs, the system will confidently deliver the wrong outcome.

The constraint has shifted.

For decades, software development was constrained by production capacity. Engineering hours were expensive. Building systems required large teams and long timelines. That cost forced organizations to think carefully about requirements.

Now the marginal cost of producing code is collapsing. Small teams can ship what once required entire departments. The friction of implementation is disappearing.

When production becomes cheap, clarity becomes scarce.

Recent research on AI generated code shows a rise in logic errors rather than syntax issues. The code compiles. It runs. But it may not reflect the intended business logic. At the same time, Google's DORA metrics indicate rising bug rates and increased review time as AI adoption grows.

This is not because AI cannot code. It is because we are not specifying well enough.

Amazon recognized this shift and launched AWS Cairo, an environment that requires developers to define testable specifications before generating code. In other words, you must state how success will be measured before the system writes anything.

That design decision reflects a deeper truth. The bottleneck is no longer the programmer's typing speed. It is the clarity of the architect's thinking and the product manager's intent.

And increasingly, those roles are converging.

In the AI era, programmer, architect, and product manager are merging into a single high leverage role.

The most valuable individuals combine:

- Deep domain knowledge
- Strong system architecture skills

- Clear understanding of the tech stack
- Ability to write precise, testable specifications
- Judgment to evaluate whether outputs solve real problems

This is not about learning to code. It is about learning to specify.

Specification is a discipline. Engineers have long used acceptance criteria, test driven development, and design documents to reduce ambiguity. Now that discipline must spread beyond engineering. Marketers defining campaigns. Analysts defining models. Operations leaders defining workflows. Everyone must articulate intent clearly enough for machines to execute.

The economic implications are significant.

When code production costs approach zero, demand for software expands dramatically. Every business process that was too expensive to automate becomes viable. Regional hospitals, small manufacturers, and local governments can now build custom systems affordably.

That expansion creates opportunity. But it also raises the bar.

Low leverage work, where AI is used as simple autocomplete, will become commoditized. High leverage work, where humans orchestrate multiple agents, design architectures, and define constraints precisely, will command disproportionate value.

The dividing line is economic output per unit of human judgment.

The good news is that judgment and specification are learnable skills. Teams can practice writing clear acceptance criteria. Leaders can demand measurable success metrics. Individuals can experiment with agent workflows and observe where ambiguity causes failure.

Future directions for organizations include:

- Embedding specification training into leadership programs.
- Standardizing spec first development processes.

- Investing in AI literacy focused on orchestration, not just tool usage.
- Reducing coordination overhead by building transparent internal systems.
- Measuring value creation directly rather than rewarding activity.

We are likely in the early phase of a productivity J curve. Adoption may initially feel messy. But companies that master specification driven development will achieve dramatic productivity gains.

The core shift is simple to state but hard to execute.

AI is a super coder. Humans must become super specifiers.

The question every knowledge worker should ask is this:
Am I optimizing for writing code, or am I optimizing for defining intent clearly enough that machines create real value?

The answer will shape careers in the decade ahead.

Artificial Intelligence | Future | Programming | Jobs | Product Manager

Following

Published in Pune AI Community

10 followers · Last published just now

The Pune AI Community (PAIC) is a collaborative platform dedicated to making AI knowledge accessible from classes to masses. We bring together AI engineers, researchers, students, startups, and professionals to learn, discuss, build, and grow together.

Edit profile

Written by Yogesh Haribhau Kulkarni (PhD)

1.8K followers · 2.1K following

PhD in Geometric Modeling | Google Developer Expert (Machine Learning) | Top Writer 3x (Medium) | More at https://www.linkedin.com/in/yogeshkulkarni/